



Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

WBL-25-007

March 3, 2025

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License No. NPF-96
NRC Docket No. 50-390, 50-391

Subject: **Licensee Event Report 390/2025-001-00, Loss of The Control Room
Emergency Air Temperature Control System Due to Chiller Oil Migration**

Pursuant to the reporting requirements of 10 CFR 50.73, attached is the subject Licensee Event Report concerning the Loss of The Control Room Emergency Air Temperature Control System Due to Chiller Oil Migration for Watts Bar Nuclear Plant, Unit 1, which occurred on January 4, 2025.

There are no new regulatory commitments contained in this letter. Please direct any questions concerning this matter to Jonathan Johnson, WBN Licensing Manager, at jtjohnson0@tva.gov.

Respectfully,

A handwritten signature in black ink, appearing to read "William C. Reneau", is written over a horizontal line.

William C. Reneau
Site Vice President
Watts Bar Nuclear Plant

U.S. Nuclear Regulatory Commission
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March 3, 2025

Enclosure: LER 390/2025-001-00, "Loss of The Control Room Emergency Air Temperature Control System Due to Chiller Oil Migration"

cc (w/Enclosure):

NRC Regional Administrator – Region II
NRC Senior Resident Inspector – Watts Bar Nuclear Plant
NRC Project Manager – Region II

ENCLOSURE
Tennessee Valley Authority
Watts Bar Nuclear Plant
Unit 1

**LER 390/2025-001-00, "Loss of The Control Room Emergency Air Temperature Control
System Due to Chiller Oil Migration"**



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number

1. Facility Name

Watts Bar Nuclear Plant, Unit 1

☒ 050
☐ 0522. Docket Number
00390

3. Page

1 OF 6

4. Title

Loss of The Control Room Emergency Air Temperature Control System Due to a Chiller Oil Migration

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
01	04	25	2025	- 001 -	00	03	03	25	Watts Bar Nuclear Plant, Unit 2		00391
									Facility Name N/A	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact

Kevin Lovell, Licensing Engineer

Phone Number (Include area code)

423-365-8196

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☒ No☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month

Day

Year

N/A

N/A

N/A

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On January 4, 2025 at 0855 EST, with Unit 1 and Unit 2 at 100% rated thermal power, both trains of the Control Room Emergency Air Temperature Control System (CREATCS) were simultaneously inoperable. This condition occurred when the A-A Main Control Room (MCR) chiller experienced an oil migration event and tripped on oil failure while the B train of CREATCS had been previously declared inoperable for a B-B MCR chiller maintenance outage. The A train of CREATCS was later restored at 1408 EST on January 4, 2025 following the addition of MCR chiller oil, adjustment of the chiller's temperature control valve (TCV) and subsequent observation of stable chiller parameters. Applicable technical specification (TS) limiting conditions for operability (LCO) were entered for less than six hours as a result.

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Watts Bar Nuclear Plant, Unit 1	☒ 050	2. DOCKET NUMBER 00390	3. LER NUMBER				
	☐ 052		<table border="1"><thead><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr></thead><tbody><tr><td>2025</td><td>- 001</td><td>- 00</td></tr></tbody></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2025
YEAR	SEQUENTIAL NUMBER	REV NO.					
2025	- 001	- 00					

NARRATIVE**I. Plant Operating Conditions before the Event**

Watts Bar Unit 1 and Unit 2 were at 100 percent Rated Thermal Power (RTP) before the event.

II. Description of Event**A. Event Summary**

On January 4, 2025 at 0855 EST, with Unit 1 and Unit 2 at 100% rated thermal power, both trains of the Control Room Emergency Air Temperature Control System (CREATCS){EILS:VI} were simultaneously inoperable. This condition occurred when the A-A Main Control Room (MCR) chiller experienced an oil migration event and tripped on oil failure while the B train of CREATCS had been previously declared inoperable for a B-B MCR chiller maintenance outage. The A train of CREATCS was later restored at 1408 EST on January 4, 2025 following the addition of MCR chiller oil, adjustment of the chiller's temperature control valve (TCV){EILS:TCV}, and subsequent observation of stable chiller parameters. Applicable technical specification (TS) limiting conditions for operability (LCO) were entered for less than six hours as a result.

This condition is being reported to the Nuclear Regulatory Commission (NRC) in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

The B-B MCR chiller had been removed from service for a maintenance outage when the A-A MCR chiller trip occurred.

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Watts Bar Nuclear Plant, Unit 1

☒ 050☐ 052**2. DOCKET NUMBER**

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YEAR

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- 001

REV
NO.

- 00

NARRATIVE**C. Dates and approximate times of occurrences**

Dates and Approximate Times	Occurrence
01/04/25 08:55	A-A MCR Chiller was found tripped with Oil Failure Light lit while B-B MCR Chiller was out of service for outage
01/04/25 09:50	Assistant Unit Operator (AUO) reported that the A-A MCR chiller restarted on its own with no operator action.
01/04/25 12:50	Mechanical maintenance group (MMG) reported that oil had been added to the A-A MCR chiller to 3/4 full in the sight glass. All chiller parameters were running stable and as expected.
01/04/25 13:30	Engineering provided an evaluation of the A-A MRC chiller condition to operations, supporting operability. The A-A MCR chiller had been operating as expected.
01/04/25 14:08	Units 1 and 2 exited the applicable TS LCO's.

All times are in EST

D. Manufacturer and model number of each component that failed during the event

N/A

E. Other systems or secondary functions affected

None.

F. Method of discovery of each component or system failure or procedural error

Plant personnel investigating the A-A MCR chiller trip determined that the chiller oil level was low.

G. The failure mode, mechanism, and effect of each failed component

The most likely cause of the A-A MCR chiller trip was a low oil level within the sump due to oil migration to the chiller's evaporator. Oil migrated during low load conditions which produced low compressor discharge pressures and an overall low refrigerant flow velocity. The low sump oil level caused a loss of oil pressure and a subsequent oil failure trip.

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NARRATIVE**H. Operator actions**

Operators ensured that the chiller restarted with maintenance personnel present to perform oil additions and necessary adjustments.

I. Automatically and manually initiated safety system responses

There were no automatic or additional manual safety system responses.

III. Cause of the event**A. Cause of each component or system failure or personnel error**

The most likely cause of the A-A MCR chiller trip was a low oil level within the sump due to oil migration to the chiller's evaporator. Oil migrated during low load conditions which produced low compressor discharge pressures and an overall low refrigerant flow velocity. The low sump oil level caused a loss of oil pressure and a subsequent oil failure trip.

B. Cause(s) and circumstances for each human performance related root cause

No human performance issues were attributed to this event.

IV. Analysis of the event

The CREATCS provides temperature control for the control room following isolation of the control room. The CREATCS consists of two independent and redundant trains.

On January 4, 2025 at 0855 EST both trains of the CREATCS were simultaneously inoperable. This condition occurred when the A-A Main Control Room (MCR) chiller experienced an oil migration event and tripped on oil failure while the B train of CREATCS had been previously declared inoperable for a B-B MCR chiller maintenance outage. The A train of CREATCS was later restored operable at 1408 EST after maintenance personnel added oil to the chiller system, adjusted the chiller's Metrex TCV, and operators observed stable parameters. Both Trains of MCR temperature control (CREATCS) were lost for less than six hours. The temperature within the MCR was maintained below the design limit.

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NARRATIVE**V. Assessment of Safety Consequences**

Both Trains of MCR temperature control (CREATCS) were lost for less than six hours. The temperature within the MCR was maintained below the design limit of 104 degrees Fahrenheit.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

There are no other systems that would perform the function of CREATCS.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not Applicable.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

The A train CREATCS was returned to service in approximately 5.22 hours.

VI. Corrective Actions

This event was entered into the TVA Corrective Action Program and is being tracked under Condition Report 1982838.

A. Immediate Corrective Actions

Oil was added to the MCR chiller system and the associated Metrex TCV was adjusted to maintain a discharge pressure between 225-235 psig.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

Operator rounds instructions were updated with new monitoring values to ensure the discharge pressure of the chiller system is maintained within acceptable bands. Associated chiller preventive maintenance instruction revisions are planned to update discharge pressures. Systems Engineering is also monitoring chiller oil levels and discharge pressures to ensure further TCV adjustments are not required.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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	☐ 052		YEAR 2025	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE

VII. Previous Similar Events at the Same Site

WBN LER 391-2025-001-00 documented a loss of both trains of CREATCS on November 22, 2024 due to a MCR Chiller TCV malfunction.

VIII. Additional Information

None.

IX. Commitments

None.